

JONGHA KIM

jonghakim@korea.ac.kr ♦ [Website](#) ♦ [LinkedIn](#) ♦ [Google Scholar](#)

RESEARCH INTERESTS

I develop reliable and efficient multimodal large language model (MLLM) systems that retrieve relevant evidence, reason over complex long-context inputs, and coordinate specialized agents and tools. My research spans MLLM post-training, evidence and token selection, and agentic workflows for practical, scalable deployment.

EDUCATION

Korea University

M.S. & Ph.D. Integrated Student in Computer Science and Engineering
Advisor: Prof. [Hyunwoo J. Kim](#), MLV Lab
GPA: 4.15/4.5

Seoul, Republic of Korea
Sep 2022–Aug 2028 (Expected)

Korea University

B.S. in Computer Science and Engineering; B.S. in Statistics (Double Major)
GPA: 4.22/4.5 (Major: 4.25/4.5)

Seoul, Republic of Korea
Mar 2018–Aug 2022

EXPERIENCE

Adobe Research

Research Scientist Intern, Video AI Lab

Mentors: Dr. [Joon-Young Lee](#), Dr. [Seoung Wug Oh](#), and Dr. [David Seunghyun Yoon](#)

San Jose, CA, USA
May 2026–Sep 2026

- Designed an MLLM-based agentic workflow and a large-scale pseudo-labeling pipeline for referring video matting, improving supervision quality and scalability.

MLV Lab

M.S. & Ph.D. Integrated Student
Advisor: Prof. [Hyunwoo J. Kim](#)

Seoul, Republic of Korea
Sep 2022–Current

MLAI Lab

Undergraduate Research Intern
Advisor: Prof. [Sung Ju Hwang](#)

Seoul, Republic of Korea
Jan 2021–Jul 2022

Upstage AI

Teaching Assistant & Mentor, BoostCamp AI Tech Cohort 1–3
Collaborated with Prof. [Tae Hyun Oh](#)

Seongnam, Republic of Korea
Dec 2020–Jun 2022

VoyagerX

Intern

Seoul, Republic of Korea
Jul 2020–Jan 2021

- Curated training data, optimized and deployed TensorFlow Lite (TFLite) on-device models for [vFlat](#), a real-time mobile document scanner with 10M+ downloads on Google Play Store.

SW Maestro

10th Trainee, SW Maestro Program

Seoul, Republic of Korea
May 2019–Nov 2019

- Developed a computer vision model for estimating apple sugar content, leading to a registered Korean patent and a national big-data startup competition award.

PUBLICATIONS

C: Conference, J: Journal, W: Workshop, U: Under Review; * equal contribution

- [C7] **Context Blindness in DPO: Mitigating Object Hallucination in MLLMs via Context-Calibrated Preference Optimization**
Byungoh Ko, Jinyoung Park, [Jongha Kim](#), Jeehye Na, Jaewon Cho, Hyunwoo J. Kim
European Conference on Computer Vision (ECCV), 2026

- [C6] **VideoSearch-R1: Iterative Video Retrieval and Reasoning via Soft Query Refinement**
Seohyun Lee*, Seoung Choi*, Dohwan Ko*, Jongha Kim, Hyunwoo J. Kim
European Conference on Computer Vision (*ECCV*), 2026
- [C5] **DocPrune: Efficient Document Question Answering via Background, Question, and Comprehension-aware Token Pruning**
Joonmyung Choi, Sanghyeok Lee, Jongha Kim, Sehyung Kim, Dohwan Ko, Jihyung Kil, Hyunwoo J. Kim
In collaboration with [Adobe](#) Research
IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR*), 2026
- [C4] **Relevance-aware Multi-context Contrastive Decoding for Retrieval-augmented Visual Question Answering**
Jongha Kim, Byungoh Ko, Jeehye Na, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
IEEE/CVF Winter Conference on Applications of Computer Vision (*WACV*), 2026
- [C3] **TabFlash: Efficient Table Understanding with Progressive Question Conditioning and Token Focusing**
Jongha Kim, Minseong Bae, Sanghyeok Lee, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
AAAI Conference on Artificial Intelligence (*AAAI*), 2026
- [C2] **VidChain: Chain-of-Tasks with Metric-based Direct Preference Optimization for Dense Video Captioning**
Ji Soo Lee*, Jongha Kim*, Jeehye Na, Jinyoung Park, Hyunwoo J. Kim
AAAI Conference on Artificial Intelligence (*AAAI*), 2025
- [C1] **Groupwise Query Specialization and Quality-Aware Multi-Assignment for Transformer-based Visual Relationship Detection**
Jongha Kim*, Jihwan Park*, Jinyoung Park*, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim
IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR*), 2024
- [J1] **Improved Query Specialization for Transformer-based Visual Relationship Detection**
Jongha Kim, Jihwan Park, Jinyoung Park, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim
Information Sciences, 2026
- [W2] **Concept Bottleneck with Visual Concept Filtering for Explainable Medical Image Classification**
Injae Kim*, Jongha Kim*, Joonmyung Choi, Hyunwoo J. Kim
MedAGI Workshop at International Conference on Medical Image Computing and Computer-Assisted Intervention (*MedAGI@MICCAI*), 2023 (**Oral Presentation**)
- [W1] **Object Detection in Aerial Images with Uncertainty-Aware Graph Network**
Jongha Kim, Jinheon Baek, Sung Ju Hwang
VOLI Workshop at European Conference on Computer Vision (*VOLI@ECCV*), 2022
- [U3] **SuperClip Pyramid for Video Temporal Grounding**
Sanghyeok Lee, Juyeon Ko, Joonmyung Choi, Jongha Kim, Dohwan Ko, Ji Soo Lee, Hyunwoo J. Kim
In collaboration with [KT](#) Gen AI Lab
Under Review
- [U2] **CEPO: Coordinator-Executor Prompt Optimization for Multi-Agent Document Understanding**
Jongha Kim*, Seohyun Lee*, Jonathan Minh Chung, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
Under Review
- [U1] **Reason in the Words You Speak: Idiolectal Paraphrasing Off-Policy Traces for Reasoning Distillation in VideoLLMs**
Ji Soo Lee, Jinyoung Park, Seohyun Lee, Jongha Kim, Joonmyung Choi, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
Under Review

PATENTS

Method and apparatus for dense video captioning based on chained task learning and metric loss for fine-grained understanding of video language model

Hyunwoo Kim, Ji Soo Lee, Jongha Kim, Jeehye Na, Jinyoung Park

Korea Patent No. 10-2025-0164550.

Information providing method and system for sharing fruit information including sugar content information measured through image vision processing

Sanghoon Lee, Hyemin Song, Jongha Kim, Hyun Kim
Korea Patent No. 10-2020-0153010.

Method for measuring sugar content of apple using image

Sanghoon Lee, Hyemin Song, Jongha Kim
Korea Patent No. 10-2242155, registered on Apr 14, 2021.

HONORS & AWARDS

Travel Grant for CVPR 2024 , CVPR 2024	Jun 2024
Graduate School Outstanding Freshman Scholarship , Korea University	Sep 2022
Dean's List , Korea University	Spring 2019
Semester High Honors , Korea University	Fall 2018/2019/2021, Spring 2019/2020/2021
2020 Agrifood Public Big Data Startup Competition , Rural Development Administration	Aug 2020
National Scholarship for Science and Engineering , Korea Student Aid Foundation	Spring 2018–Spring 2022

ACADEMIC SERVICES

Reviewer of **NeurIPS** (2024–2026), **CVPR** (2024–2026), **ICLR** (2025–2026), **AAAI** (2025–2027), **AISTATS** (2025–2026), **ICML** (2025), **WACV** (2026–2027), and **MedAGI@MICCAI** (2024–2026)

TEACHING & TALKS

Teaching Assistant , Special Topics in Artificial Intelligence (AAA740), Korea University	Fall 2024
<i>Korea University</i>	Seoul, Republic of Korea
Korea University & LG AI Workshop , Korea University	Feb 2023
<i>Korea University</i>	Seoul, Republic of Korea
Defining and Solving Problems in Deep Learning Projects	Nov 2021
<i>BoostCamp AI Tech 2nd Course, Upstage AI</i>	Virtual

SKILLS

Language: Korean (native); English (fluent)

Programming: Python, PyTorch, Git, Bash (most proficient); TensorFlow, SQL, Node.js, React.js, Vue.js (experienced)